Importance Resampling and MCMC: Recap

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Known up to a constant

$$f(x) = c\, f^*(x), \qquad c = \frac{1}{\int f^*(x)\, dx}$$

- f^* is easy to evaluate, but the integral may not be.
- Today's two samplers use only f^*.

	needs an envelope g?	draws
Importance resampling	yes	resampled from g's candidates
Markov chain Monte Carlo	no	a dependent sequence

Importance resampling

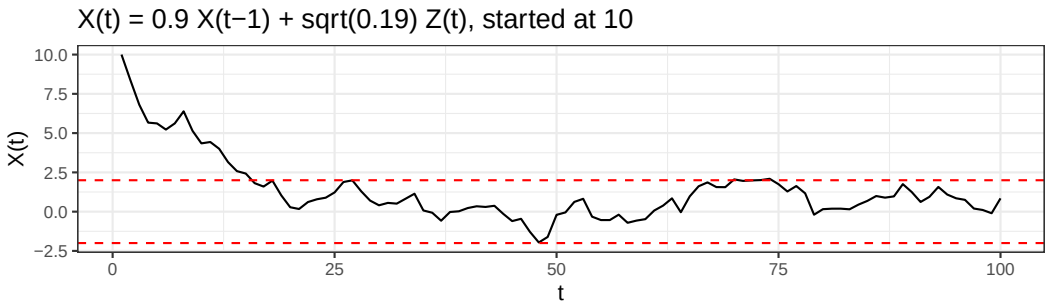
1. Draw $Y_1, \ldots, Y_m$ from g.
2. Weight each by $f^*(Y_i)/g(Y_i)$, and normalize the weights to sum to 1.
3. Resample with `sample(y, replace = TRUE, prob = w)`.

- c **cancels** when the weights are normalized.
- Nothing is rejected. The cost is a small bias that shrinks as m grows.
- g must cover f, with tails at least as heavy. Otherwise a few candidates carry almost all the weight.

Example: $Z \mid Z \geq 4.5$ with candidates $4.5 + \text{Exp}(1)$.

Markov chains and burn in

A **Markov chain** draws $X(t)$ given only $X(t-1)$. If f is **stationary** and the chain can reach all of f's support, averages along the chain converge to $E_f(h(X))$.



The early draws are not from f. Dropping them is **burn in**.

Metropolis-Hastings

Accept-reject, with a candidate that depends on where the chain is.

1. Propose $X^* \sim g(\cdot \mid X(t-1))$.
2. Move to X^* if
$$U \leq \frac{f^*(X^*)}{f^*(X(t-1))} \frac{g(X(t-1) \mid X^*)}{g(X^* \mid X(t-1))},$$
and otherwise stay: $X(t) = X(t-1)$.

- f appears only in a ratio, so c cancels.
- A move to lower density is sometimes accepted. That is what makes it a sampler and not an optimizer.
- Check with a **trace plot**, and with a QQ plot when Q is known.

Gibbs sampling

The **full conditional** of x_i is its distribution given all the other coordinates. As a function of x_i it is proportional to f, so drop the factors without x_i.

- Step through the coordinates, drawing each from its full conditional given the **current** values of the rest.
- A special case of Metropolis-Hastings in which every candidate is accepted.
- Needs full conditionals we can draw from directly, and needs burn in like any chain.

Bivariate normal, correlation 0.8: $X_1 \mid X_2 = t \sim N(0.8\,t,\ 0.36)$.

Bayes and the Challenger

$$\pi(\theta \mid x) \propto f(x \mid \theta)\, p(\theta)$$

The posterior is known up to a constant, so all three samplers apply.

Logistic model for O-ring distress against launch temperature, 23 flights:

- Metropolis-Hastings gives $P(\text{distress at } 31^\circ\text{F})$ between 0.88 and 1.00 (95%), and a grid agrees.
- Every flight was at least 22°F warmer, so the answer leans on the model's form outside the data.

Failure case. Importance resampling with the prior as the envelope: the prior is far wider than the posterior, so a few candidates carry nearly all the weight. The effective sample size is about 540 out of 100,000.